led to the development of sunscreens which use nanoparticles of zinc oxide manufactured as a completely clear liquid instead of the familiar thick white paste. Some other products, for example specific moisturizers with certain actives, may use nanosomes (little pocket-like nanoparticles which disintegrate or melt upon application to skin) to expedite the moisturizing effect.

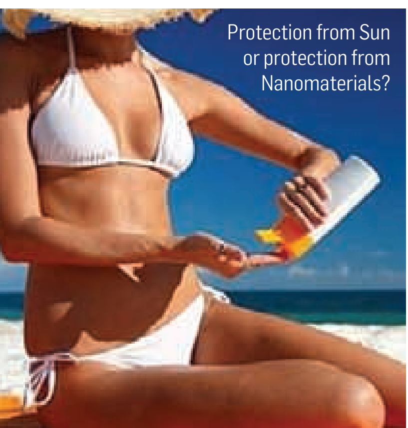
So what is the flipside?

It stands to reason that any substance which can penetrate skin may also make its way into the blood stream and increase risk of disease and reactions. Groups such as Friends of the Earth have expressed concern about their physical properties and their potential risk to cells and organs.

Fiend of Earth published a report entitled "Nanomaterials, Sunscreens and Cosmetics: Small Ingredients, Big Risks" a few years back and since then they have had come down hard on this subject. In the items they reviewed, they found several face creams that list carbon fullerenes as their ingredients- a substance found to cause brain damage in fish and toxic effects in human liver cells.

Manufactured nanomaterials used in sunscreens (such as zinc oxide and titanium oxide) can

1. **Damage human colon cells:** A study from the University of Utah showed that nanoparticles of zinc oxide are toxic to colon cells even in small amounts. In nano scale ZnO particles were around twice as potent as larger ZnO particles in their ability to kill these cells under idealized conditions.
2. **Damage brain stem cells in mice:** A study from China found that zinc oxide nanoparticles can damage the brains of mice by killing important brain stem cells. In another study, Japanese scientists injected pregnant mice with nano titanium dioxide and recorded changes in gene expression in the brains of their fetuses. These changes have been associated with autistic disorders, epilepsy and Alzheimer's disease.



3. **Travel up the food chain from smaller to larger organisms:** A study by researchers at Arizona State University, the Georgia Institute of Technology, and Tsinghua University in China found through a dietary experiment that Daphnia (a "water flea" that provides important nutrition for aquatic life) can transfer nano titanium dioxide to larger organisms (in this case Zebrafish).